

Python Placement Training - Day 3 Notes

reversed() Function

reversed() traverses elements in reverse order without changing the original list.

Example

```
for i in reversed(l):  
    print(i)
```

Reverse Traversal using range()

range(len(digits)-1, -1, -1) traverses from last index to first index.

Why -1 is Used

range() stops before stop value.
Using -1 ensures index 0 is included.

for i in nums

Loop variable stores elements directly.

for i in range(len(nums))

Loop variable stores indexes.

count() Logic

nums.count(nums[i]) checks frequency of current element.

Problem with Wrong Logic

count(i) checks occurrence of indexes instead of actual elements.

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Using count() inside loop results in $O(n^2)$ complexity.

Correct count() Solution

```
for i in range(len(nums)):
    if nums.count(nums[i]) < 2:
        return nums[i]
```

XOR Solution

```
ans = 0
for i in nums:
    ans ^= i
```

Important XOR Properties

$a \oplus a = 0$
 $0 \oplus a = a$

XOR Logic

Duplicate numbers cancel each other and only the single occurring number remains.

Time Complexity

count() solution $\rightarrow O(n^2)$
XOR solution $\rightarrow O(n)$

Left Rotation

Left rotation is anti-clockwise rotation.

Right Rotation

Right rotation is clockwise rotation.

Rotation Example

Original: [1,2,3,4,5]
Left Rotate: [2,3,4,5,1]
Right Rotate: [5,1,2,3,4]